

RESEARCH ARTICLE

Thinning effects on forest production and rainfall redistribution: Reduced soil water deficit and improved sustainability of semiarid plantation forestlands

Ling-Chao Meng^{1,2,3} | Zeng Cui^{1,2,3} | Ze Huang^{1,2} | Manuel Esteban Lucas-Borja⁴  | David Dunkerley⁵ | Gao-Lin Wu^{1,2,3,6} 

¹State Key Laboratory of Soil Erosion and Dryland Farming on the Loess Plateau, Institute of Soil and Water Conservation, Chinese Academy of Sciences and Ministry of Water Resource, Yangling, PR China

²Institute of Soil and Water Conservation, Northwest A&F University, Yangling, PR China

³University of Chinese Academy of Sciences, Beijing, PR China

⁴Escuela Técnica Superior Ingenieros Agrónomos y Montes, Universidad de Castilla-La Mancha, Albacete, Spain

⁵School of Earth Atmosphere and Environment, Monash University, Melbourne, Victoria, Australia

⁶CAS Center for Excellence in Quaternary Science and Global Change, Xi'an, China

Correspondence

Gao-Lin Wu, State Key Laboratory of Soil Erosion and Dryland Farming on the Loess Plateau, Northwest A&F University, No. 26 Xinong Road, Yangling, Shaanxi 712100, PR China.
Email: wugaolin@nwsuaf.edu.cn

Funding information

National Natural Science Foundation of China, Grant/Award Numbers: NSFC41977063, NSFC41930755, NSFC41722107; the Strategic Priority Research Program of the Chinese Academy of Sciences, Grant/Award Number: XDB40000000

Abstract

Large-scale planted afforestation worldwide has exacerbated soil desiccation and led to a serious shortage of water resources for forests sustainability in semiarid areas. Consequently, thinning is discussed as an important management approach to conserve water resources and to maintain plantation forestland sustainability. Here, we explored the effects of thinning intensity on semiarid forest production and soil water supply. Three artificial forests of *Populus cathayana* with an age of 15 yr were selected and thinned with different intensities [1/3 amount of trees removed ($T_{1/3}$), 2/3 amount of trees removed ($T_{2/3}$), and using no thinning (NT) as a control]. Results showed that thinning presented a significant impact on the above-ground three-dimensional structure of planted forests and increased forest gaps. Meanwhile, the rainfall redistribution pattern changed significantly. Thinning increased throughfall ($T_{1/3}$, $43\% \pm 9\%$; $T_{2/3}$, $79\% \pm 18\%$), reduced stemflow ($T_{1/3}$, $-44 \pm 2\%$; $T_{2/3}$, $-71\% \pm 2\%$), and increased the soil water replenishment ($T_{1/3}$, $34\% \pm 7\%$; $T_{2/3}$, $64\% \pm 19\%$) of forest land. Meanwhile, the dead root channel caused by thinning effectively increased the water infiltration rate and groundwater replenishment, and effectively reduced soil water deficit ($T_{1/3}$, $41\% \pm 33\%$; $T_{2/3}$, $57\% \pm 33\%$). Through trade-off analysis, we found that non-thinning aboveground biomass yields were the highest, and $T_{2/3}$ yields higher water yield, while $T_{1/3}$ can maintain acceptable aboveground biomass while keeping better soil water. We conclude that thinning increased soil water supply, reduced soil water deficit and maintained forestland sustainability, and there are synergies of thinning on forest production and soil water supply through rainfall redistribution for maintaining the sustainability of semiarid plantation forestlands.

KEYWORDS

forest production, forest thinning management, rainfall redistribution, soil water replenishment, sustainability

1 | INTRODUCTION

Afforestation has become an important measure for countries around the world to deal with climate change (IPCC, 2014). The assessments showed that from 2010 to 2015, global forest cover increased by 2.5 million ha yr⁻¹ (FAO, 2015). The increase in forest coverage attracted more and more attention because of its many benefits to human well-being and the environment (Liu et al., 2018; Stanturf & Mansourian, 2020), such as desertification control (Wang et al., 2010), soil water conservation (Chirino et al., 2006), and carbon capture (Elbakidze et al., 2011). However, in recent years, the increase of global vegetation browning area and the stabilization of primary productivity indicate that soil water deficit has great restrictions on greening earth (Feng et al., 2021).

In China, which accounts for about 1/3 of the world's plantation area (6.93×10^5 km²), there is a serious problem of soil water deficit and desiccation (Cao et al., 2011; Deng et al., 2016; Jia et al., 2017). The Chinese government has implemented a series of large-scale afforestation projects (e.g., the Grain-for-Green Program) (Cao et al., 2011), and great progress has been obtained in afforestation. These afforestation projects are mainly concentrated in arid and semiarid areas of China (Cao et al., 2011). However, large-scale afforested lands undoubtedly exacerbated the degree of soil dryness in the arid and semiarid areas (Deng et al., 2016). More studies have shown that the warming rate in the arid and semiarid regions are more significant than that in other regions, and the rapid climate warming and drying will make the region face greater potential risks (Huang, Yu, et al., 2017). Zastrow (2019) found that arid areas in China had increased by roughly 1.6×10^7 km², which is about equal to the land area of Iran, and may be due largely to anthropogenic climate change since 1980. A study in 2016 found that the restored forest ecosystem can intercept rainfall and reduce surface runoff (Feng et al., 2016). Dry climate exacerbates this situation and leads to human water shortages (Feng et al., 2016; Ge et al., 2020).

Reducing forest density (i.e., thinning) as a potential strategy can improve the hydrological balance by improving forest health and increasing soil infiltration and runoff (del Campo et al., 2019). It has been observed that thinning can improve drought resistance and drought resilience by reducing competition (such as water, light, and nutrients) (Giuggiola et al., 2013; Olivar et al., 2014). However, some studies showed that thinning can inhibit the growth of trees by changing the microclimate conditions, leading to asymmetric growth of trees (Guillemot et al., 2015; Jimenez et al., 2011). The reasons for the different response effects of thinning may be related to thinning methods, thinning intensity, stand density, forest age, site climatic conditions, and time after thinning (del Campo et al., 2014; Sohn et al., 2016). Therefore, for arid and semiarid areas, the effect of conventional thinning methods on plantations needs to be further clarified. Meanwhile, other policy-related issues remain unresolved, such as the optimal thinning intensity and its impact duration. To answer these questions, it is necessary to conduct long-term research on forestland with different thinning intensities. Though some studies have been reported on the thinning of semiarid plantation forestland, knowledge of soil water of afforested land based on the change of aboveground structure and rainfall redistribution remains essentially

lacking. Progressively after thinning, the root systems decay, forming root macropores (Wu et al., 2021). In turn, this may improve soil structure (Wu et al., 2019), promote rainfall infiltration (Wu et al., 2016; Wu et al., 2021), and restore soil water conditions (especially the water supplement of deep soil layers) (Bogner et al., 2010), but this part is always ignored. The objectives of our study are therefore to: (1) explore the effects of different thinning intensities on three-dimensional structures aboveground and the pattern of rainfall redistribution; (2) determine the reasonable thinning intensity for semiarid plantation forestlands management by studying the effects of forest thinning on the soil water storage and the trade-off between soil water and aboveground biomass.

2 | MATERIALS AND METHODS

2.1 | Study site

The experiment was conducted near Lvliang City (37°19' N, 111°05' E, 715 m) and Jinzhong City (37°14' N, 112°06' E, 741 m) in Shanxi Province, China (Figure 1). The average altitude of the study area is 574–2567 m, belonging to a semiarid temperate continental climate. The mean annual temperature and the mean annual precipitation are 10.1 °C and 457 mm, respectively. The rainfall in July and August accounts for more than 60% of the total annual rainfall (Huang et al., 2020). The soils in the study area are Loessial and Castanozems (Liu et al., 2018).

2.2 | Experimental design and sampling

The studied *P. cathayana* afforested lands were planted on three farm-lands in 2000 (Plot 1) and 2005 (Plot 2 and 3) as part of the main policy of the Grain-for-Green Program (the density range of local conventional planting from 1600 to 1800 hm⁻²). Since the establishment of the afforested land, local farmers regularly managed and maintained the afforested land. They collect the ground litter to meet their living needs such as raising livestock, cooking, and heating. Therefore, there is no obvious litter covering the afforested land (Figure 1). In 2015, the local conventional thinning method, artificial thinning, was carried out instead of large-scale mechanical thinning to reduce the disturbance to the soil, including several intensities at all sites: moderate thinning 1/3 of trees removed, $T_{1/3}$, heavy thinning (2/3 of trees removed, $T_{2/3}$), and no thinning (NT) as the control (Figure 1). All thinning measures were completed at one time in 2015. In the process of thinning, farmers selected a certain number of trees with good growth and marked them, and then used the hand saw to cut them at 10–20 cm above the ground. The above-ground part will be sent to the local wood processing plant, and the underground part (root) will be kept in place (not excavated), while the remaining trees with poor growth will continue to grow. The underground part (root) left in the ground gradually dies because of the loss of the above-ground part and the loss of nutrition supply, and then gradually decays from the centre of the root to the outside, finally forming a hollow root macropore (Figure 1). In our study, we used the two thinning

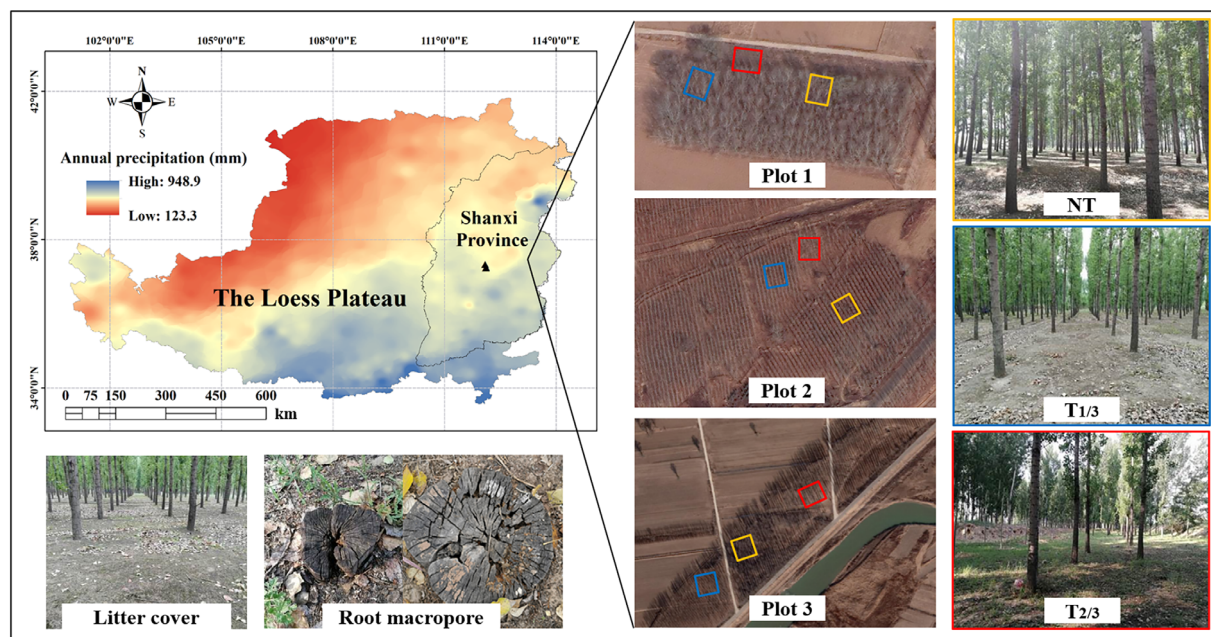


FIGURE 1 Study location, experimental layout of three plantation forestland plots, and the photos of NT, $T_{1/3}$, and $T_{2/3}$ after the thinning treatments were applied, including the litter cover and root macropore [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/ldr.4379)]

intensity treatments ($T_{1/3}$, $T_{2/3}$) as the research object and NT forest as the control. In our three experimental areas, we established three plots (50 m × 20 m), and selected three quadrats (10 m × 10 m) for each plot. A total of nine quadrats were selected for each thinning treatment ($T_{1/3}$, $T_{2/3}$). In May 2020 (before the rainfall redistribution pattern experiment), we surveyed the three-dimensional architecture of forest in all sample plots. In the three sample plots, the forest structure without thinning treatment is similar, with the mean crown density (CD) of 77%, the mean diameter at breast height (DBH) of 21.12 cm, the mean height (H) of 18.9 m, the mean under-branch height (UBH) of 3.86 m, the mean crown diameter (C) of 23.6 m; the mean lateral branch angle (LBA) of 37.76° (Table 1). For each plot, we investigated all trees in the sample plot by using the method of measuring each tree (due to man-made management measures, there are no other derived species in our test plot, which are all the trees planted at the beginning). We estimated the tree height (m) and the under-branch height (m) with an altimeter, measured DBH (cm) at 1.3 m with DBH ruler and the crown (m²) width at noon with tape, and calculated the canopy density (%) of the forest land, measured the angle of the upper side branch of the trunk with a tree goniometer in all sample plots. At the same time, we counted the number of root macropores of >10 cm in each plot. The aboveground biomass was calculated as follows (Gerwing & Farias, 2000):

$$\text{DBH} < 20\text{ cm}: \ln(\text{mass}) = -1.754 + 2.665 \ln(\text{DBH}) \quad (1)$$

$$\text{DBH} > 20\text{ cm}: \ln(\text{mass}) = -0.151 + 2.170 \ln(\text{DBH}) \quad (2)$$

where DBH is diameter at breast height, mass is the fresh aboveground biomass (Mg·hm⁻²), and the ratio of dry aboveground biomass to fresh aboveground biomass was 0.603.

Rainfall (R) was measured on an open ground 150 meters away from the experimental plots by using three rain gauges (diameter, 10 cm) placed evenly at plot 1. The measurement period was from June to July 2020, and each natural rainfall (precipitation >0.5 mm) was counted as a rainfall event, and two rainfall events with rainfall interval time of more than 2 hr were identified as two separate rainfall events. Meanwhile, three rain gauges (diameter, 10 cm) per treatment were systematically arranged along both diagonals of the square plot, placed near the trunk, in the middle of the crown, and at the intersection of the crowns of the two trees. The average receiving rainfall (mm) of the three rain gauges was used to represent the throughfall (mm) as stand-level of the plantation under each treatment. Stemflow was also measured from May to July 2020. Thus, the interception was assessed from May to July 2020. The trees in each treatment were divided into three diameter grades, and two representative standard trees were selected in each diameter grade. Therefore, each treatment contains six representative standard trees. On the trunk of standard wood 1.3 m above the ground, the smooth part of the bark was selected (if the bark was rough, it was carefully removed), with a high-density impermeable plastic pipe with a diameter of 25 mm to cut and wrap around the tree, and daub sealing, to prevent rainwater leakage; the other end of the plastic pipe was connected with a 15 L container. At the stand level of the plantation, the stemflow of each treatment was the sum of the products of the products of the number of trees in each diameter class and the average stemflow of standard trees in each diameter class, and the ratio to the plot area. The specific calculation process is as follows:

Throughfall (T_F , mm) was the average value of rainfall collected by throughfall gauges at each treatment level after each rainfall event;

Stemflow (S_F , mm) was calculated as follows:

TABLE 1 Table of three-dimensional architecture under different tending thinning intensity

Treatment	Crown density (%)	Mean diameter at breast height (cm)	Mean height (m)	Mean under-branch height (m)	Mean canopy thickness (m)	Mean crown (m ²)	The mean number of root macropores	Single-crop volume/mg	Aboveground biomass(mg hm ⁻²)	Mean lateral branch angle/°
NT	77.00 ± 8.09a	21.12 ± 3.55a	18.9 ± 1.94a	3.86 ± 1.01a	15.04 ± 2.92a	23.6 ± 12.21a	—	7.01 ± 0.81a	9416.08 ± 1444.15a	31.75 ± 5.17c
T _{1/3}	63.89 ± 12.61b	21.76 ± 3.30a	18.04 ± 2.98a	4.81 ± 1.37a	13.23 ± 2.76a	24.17 ± 19.34a	525 ± 27.64b	6.58 ± 0.87a	7832.32 ± 1414.51a	45.74 ± 7.39b
T _{2/3}	47.78 ± 17.76c	22.11 ± 1.28a	17.49 ± 3.55a	3.59 ± 1.82a	13.90 ± 4.60a	27.75 ± 11.78a	1100 ± 16.67a	7.21 ± 0.35a	3476.42 ± 102.63b	53.51 ± 7.95a

Notes: NT is the control, and T_{1/3} and T_{2/3} are the approximate range of thinning intensity of plantation forestland. Values are expressed as means ±SD. The different lower-case letters mean the significant differences at different thinning intensity ($p < 0.05$).

$$S_F = N \sum_{i=1}^n \frac{C_i}{n} / A \quad (3)$$

Where: N is the total number of plants in the study plot, n is the number of standard trees, A is the plot area (m²), and C_i is the stemflow volume of the i th standard tree (ml).

Canopy interception (I_c) was calculated according to the principle of water balance:

$$I_c = P - (T_F + S_F) \quad (4)$$

Where: P is the rainfall outside the forest (mm), T_F is the throughfall at stand-level of the plantation under each treatment (mm), S_F is stemflow at stand-level of the plantation under each treatment (mm).

Soil water content of each quadrat was measured in 0–400 cm soil layers at intervals 10 cm in July 2020. Within each quadrat, three sampling points were uniformly selected along the diagonal, and each sampling point was located in the centre of the planting points of the four adjacent trees. Gravimetric soil water content (SWC, %) was measured by the drying method, that is, the ratio of the weight of water lost during drying to the weight of soil after drying. Meanwhile, a soil profile with a depth of 100 cm was excavated at the centre of each treatment gradient, and the soil bulk density was measured at every 10 cm interval with cutting rings with a volume of 100 cm³. Three replicate samples were obtained for each soil layer. Soil bulk density of the 100–400 cm soil layers were represented by the mean values of the 50–100 cm soil layers.

Soil water storage (SWS) was calculated as follows:

$$SWS = \sum_{i=10}^n D_i \times B_i \times SWC_i \times 10^{-1} \quad i = 10, 20, 30, \dots, 400 \quad (5)$$

Where: SWS is soil water storage of the 0– i cm soil layers (mm), D_i is soil depth (cm), B_i is soil bulk density (g cm⁻³), and SWC_i is the gravimetric soil water content (%).

Soil total porosity (TP, %) was calculated as follows:

$$TP = \left(1 - \frac{B_i}{ds} \right) \times 100 \quad (6)$$

Where: ds is the density of soil (g cm⁻³), which was assumed to be 2.65 g cm⁻³.

2.3 | Trade-off analysis of ecosystem functioning

Bradford and D'Amato (2012) established a computational formula for trade-offs among individual benefits, providing a powerful tool to quantify the trade-offs relationship between two or more ecosystem functions. Compared with correlation analysis, trade-offs analysis can show the uneven change rate of different ecosystem functions in the same direction (Lu et al., 2014). Before calculating the trade-off value, each ecosystem function needs to be standardized to obtain the

relative benefits of each ecosystem function by using the following formula:

$$EF_{std} = (EF_{obs} - EF_{min}) / (EF_{max} - EF_{min}) \quad (7)$$

Where: EF_{std} is the relative benefit of any ecosystem function, EF_{obs} is an observed value, EF_{max} and EF_{min} are the maximum and minimum observed value, respectively. The EF_{std} ranges from 0 to 1.

After calculating the relative benefits of each ecosystem function, the trade-off between ecosystem functions is calculated by using the following formula:

$$RMSE = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (EF_i - \bar{EF})^2} \quad (8)$$

Where: RMSE represents the mean deviation, n is the number of ecosystem functions. EF_i is the relative benefit of ecosystem function, and $\bar{EF}$ is the expected benefit of the whole ecosystem functions. In the trade-off between two ecosystem functions, $\bar{EF}$ is equal to the 1:1 line.

2.4 | Statistical analysis

Redundancy analysis (RDA) can intuitively reflect the complex relationship between the two groups of variables, and retain the contribution of environmental factors to ecosystem function (Legendre et al., 2011). RDA was conducted using R software package to analyze the relationship between the solid structure of afforested lands and ecosystem functions, taking the solid structure metrics such as crown density, DBH, height, under-branch height, canopy thickness, crown, root macropores, and lateral branch angle range were used as explanatory variables and the ecosystem functions such as aboveground biomass, throughfall, canopy interception, stem flow, soil bulk density, porosity, and soil water storage were used as response variables.

The one-way analysis of variance (ANOVA) was conducted to test the difference of three-dimensional structure, rainfall redistribution, and soil water storage across the different thinning intensities. The least significant different (LSD) test was performed to determine the significant level (at $p < 0.05$) among the treatments. All statistical analyses were performed using the IBM SPSS 20.0 software (IBM, Montauk, NY, USA).

3 | RESULTS

3.1 | Changes of the three-dimensional architecture of forest and physical properties of soil under the different thinning intensities

The variations in the three-dimensional architecture of afforested lands with thinning intensity changes are shown in Table 1. Compared with non-thinning plots, thinning significantly increased the forest gaps

($T_{1/3}$, 17.03% [the change amount after thinning, same meaning below]; $T_{2/3}$, 37.95%) (Table 1). Meanwhile, thinning made individual trees tend to be coarsely dwarfed and luxuriant (Table 1). Specifically, compared with non-thinning, the average tree height of planted forestland was reduced, among which the mean height of thinning trees was reduced by 4.57% by $T_{1/3}$ and by 7.46% by $T_{2/3}$. The DBH, mean lateral branch angle, and mean crown area increased, among which the increase of thinning was 3.01%, 44.07%, and 2.41% by $T_{1/3}$, and that of thinning was 4.70%, 68.53%, and 17.58% by $T_{2/3}$. In addition, thinning also increased the number of root macropores ($T_{1/3}$, 525; $T_{2/3}$, 1100). However, the most direct and important impact of thinning is to reduce the above-ground biomass of the planted forestland. Compared with non-thinning, the aboveground biomass of $T_{1/3}$ was reduced by 56% and the aboveground biomass of $T_{2/3}$ was reduced by 75%. Thinning also has a great impact on soil physical properties. Compared with non-thinning, soil bulk density increased gradually ($T_{1/3}$, 1.43%; $T_{2/3}$, 3.57%) and total porosity decreased ($T_{1/3}$, -1.50%; $T_{2/3}$, -3.57%).

3.2 | Changes of rainfall redistribution pattern under the different thinning intensity

During the study period from June to July 2020, there were ten rainfall events in the study area, and the cumulative rainfall was 41 mm, which is roughly the same as rainfall in recent years (Table 2). Throughfall and stemflow were simultaneously recorded during rainfall events throughout the study period. In the non-thinning forest, the canopy interception accumulated value was the largest, 24 mm, accounting for 58.7% of the total rainfall. The cumulative throughfall value was 15 mm, accounting for 36.9% of the total rainfall. The cumulative value of stemflow was the smallest (2 mm), accounting for only 4.4% of the total rainfall (Figure 2). Compared with non-thinning, the rainfall redistribution pattern of planted forests changed with the change of thinning intensity. The canopy interception accumulation value decreased with the increase of thinning intensity, and the range dropped from -24% for $T_{1/3}$ to -45% for $T_{2/3}$. The cumulative value of throughfall increased with the increase of thinning intensity, and the range increased from 43% for $T_{1/3}$ to 79% for $T_{2/3}$. The stemflow decreased by 44% for $T_{1/3}$ and by 71% for $T_{2/3}$. Through redundancy analysis, the rainfall redistribution pattern mainly has a high correlation with the number of root macropores (namely density), branch angle, and canopy density.

3.3 | Effect of thinning intensities on soil water content

With the increase of thinning intensity and soil depth, the changing trend of soil water storage was different in soil layers (Figure 3a). Compared with non-thinning, with the increase of thinning intensity ($T_{1/3}$ and $T_{2/3}$), the total soil water storage capacity of 0–400 cm increased gradually with the increase of thinning intensity, and the soil water storage capacity of about $T_{1/3}$ increased by 264 mm, the soil water storage increased by 373 mm with about $T_{2/3}$, and by 41% and 57% with NT (649 mm) (Figure 3b). The soil water storage of different

Event	Date	Gross rainfall (mm)	Duration (hr)	Rainfall intensity (mm hr ⁻¹)
1	11/6	1.70	2.00	0.85
2	16/6	3.82	12.00	0.32
3	24/6	6.49	1.30	5.00
4	29/6	1.36	1.25	1.09
5	2/7	1.91	2.50	0.76
6	4/7	3.01	2.00	1.51
7	8/7	5.43	3.45	1.57
8	10/7	3.73	2.10	1.78
9	11/7	10.10	2.80	3.61
10	16/7	3.86	1.40	2.76
Total	—	41.41	30.80	—

TABLE 2 Gross rainfall, duration, and rainfall intensity of ten natural rainfall events in this study area over the study period (June to July 2020)

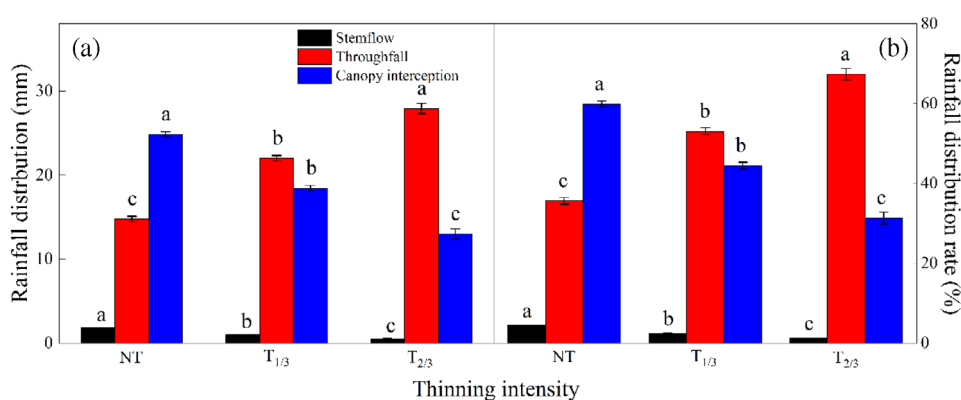


FIGURE 2 Redistribution (a) and distribution rate (b) of rainfall under different thinning intensity. NT is the control density, $T_{1/3}$ and $T_{2/3}$ are the approximate range of thinning intensity of plantation forestland. The different lower-case letters mean the significant differences within different thinning intensity ($p < 0.05$) [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/ldr.4379)]

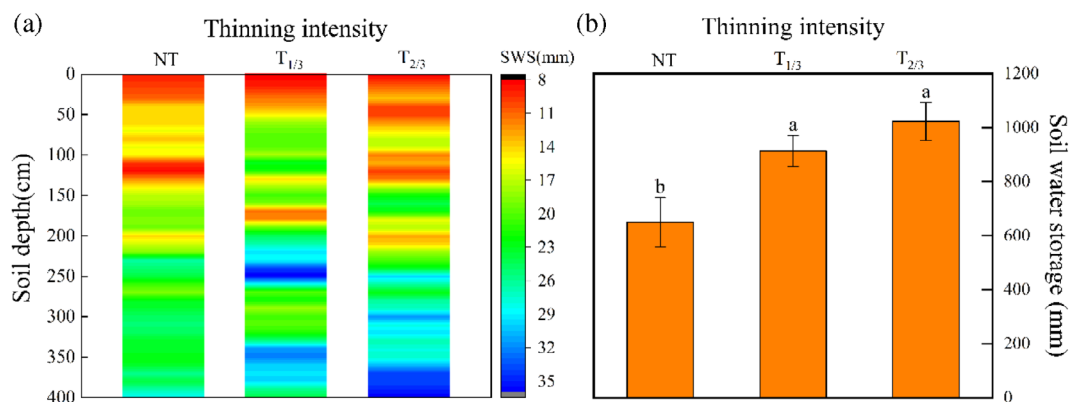


FIGURE 3 Vertical distribution (a) and collective change (b) of soil water storage under different thinning intensity. NT is the control density, and $T_{1/3}$ and $T_{2/3}$ are the approximate range of thinning intensity of plantation forestland. The different lower-case letters mean the significant differences within different thinning intensity ($p < 0.05$) [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/ldr.4379)]

thinning intensities increased obviously with the increase of soil depth, especially under $T_{1/3}$. The redundancy analysis showed that the variation of soil water storage was positively affected by canopy width, the angle of lateral branches, the thickness of canopy (plant height and the height under branches), and the change of rainfall redistribution pattern had some influence on soil water storage (Figure 4).

3.4 | Aboveground biomass and soil water content trade-offs during different thinning intensity

With the increase of thinning intensity, the aboveground biomass of the planted forestland decreased, and the soil water content increased, and the benefits of the planted forestland changed from aboveground biomass to soil water content (Figure 5). Through the

FIGURE 4 Statistical results of the redundancy analysis (RDA) under different thinning intensity. NT is the control density, and $T_{1/3}$ and $T_{2/3}$ are the approximate range of thinning intensity of plantation forestland. The red arrows indicate response variables and the blue arrows indicate explanatory variables. Thinning-TI, the mean number of root macropores-NRM, mean diameter at breast height-DBH, crown density-CD, mean crown-C, mean height-H, mean under-branch height-UBH, mean canopy thickness-CT, mean lateral branch angle-LBA, the aboveground biomass-AGB, Throughfall- T_F , Stemflow- S_F , canopy interception- C_i , soil water storage-SWS, bulk density-BD, Total porosity-TP. Acute, obtuse, and right angles between arrows indicate positive, negative, and no correlations, respectively. RDA1 and RDA2 are two vector axes of RDA. The ellipses represent 95% confidence interval [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

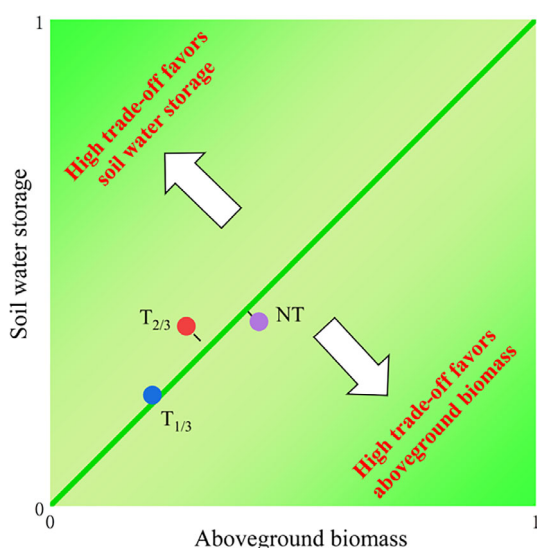
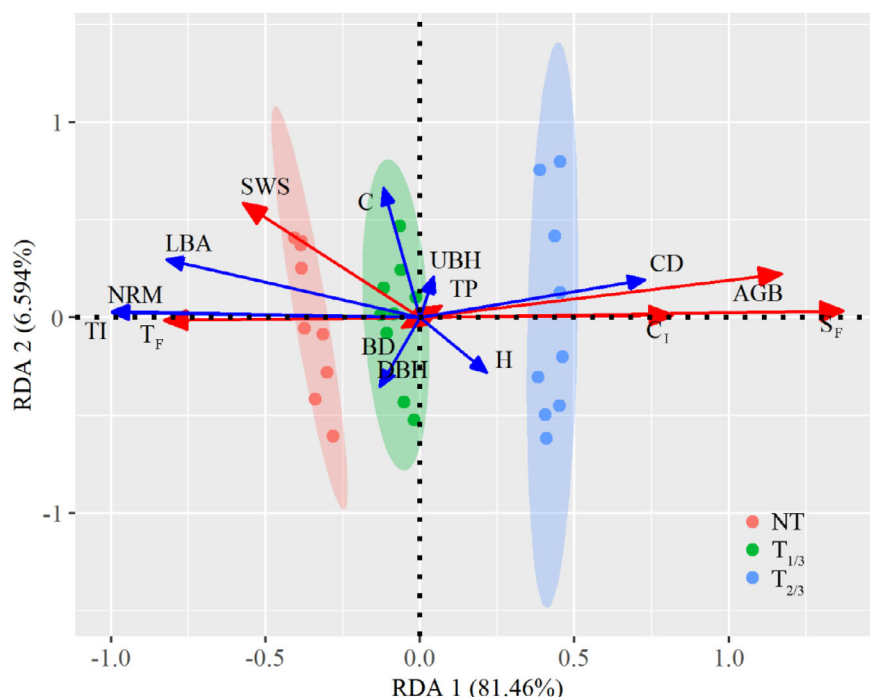


FIGURE 5 Scatter plot matrices of aboveground biomass and soil water storage functions relative benefit. The ecosystem functions relative benefit values after standardization are in the range of 0–1. And the distance of data point coordinates relative to the 1:1 line indicates which ecosystem function is more favorable under specific conditions (i.e., two arrows pointing in the direction), with the upper-left arrow indicating that the point above the 1:1 line is more toward the soil water storage and the lower-right arrow indicating that the point below the 1:1 line is more toward the aboveground biomass. NT is the control density, and $T_{1/3}$ and $T_{2/3}$ represent three thinning tending intensities, respectively [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

trade-off analysis between them, it can be concluded that the benefits of the aboveground biomass of NT-planted forestland are greater than those of the soil water storage, the benefit of above-ground

biomass of $T_{1/3}$ planted forestland could be balanced with that of soil water storage, while the benefit of aboveground biomass of $T_{2/3}$ planted forestland was less than that of soil water storage (Figure 5). In general, $T_{1/3}$ had the best eco-economic benefit compared with other treatments.

4 | DISCUSSION

Throughfall and stemflow formed by rainfall can be transported to the forest floor, which has a potentially important control effect on forest hydrology (Cheng et al., 2020; Levia & Frost, 2003; Swaffer et al., 2014). In this study, we found a relatively large interception proportion (58.7% of gross rainfall), while the actual water amount (the net rainfall [NR]) reaching the forest surface amounts for only 41.3% of gross rainfall, in which stemflow only accounted for 4.4%. Therefore, understanding the influence of the rainfall redistribution pattern of plantation forestlands is essential for maintaining the sustainability of semi-arid plantation forestlands. Forest thinning can regulate the hydrological cycle at multiple spatial and temporal scales by changing the water balance of ecosystem (Dung et al., 2012). In our study, we found that canopy interception decreased by 24% and stemflow decreased by 44% after $T_{1/3}$, canopy interception decreased by 45% and stemflow decreased by 71% after $T_{2/3}$. Our results supported the hypothesis that thinning reduced canopy interception and increased throughfall and stemflow (Limousin et al., 2008; Sun et al., 2015). The high conversion rate of gross rainfall to throughfall increases the possibility of soil water supplement (Cheng et al., 2020). However, under the same rainfall conditions, forest structure (e.g., canopy cover height, and stand density) may greatly affect the change of rainfall redistribution pattern (Sun et al., 2014; Tague et al., 2019). The results of this study showed that thinning increased the gap, improved

resource availability, increased throughfall, and increased rainfall reaching the ground. Specifically, thinning reduced competition among individuals in afforested land, increased resource availability and growth space (Niccoli et al., 2020; Park et al., 2018), and thus promoted photosynthesis of trees (Gauthier & Jacobs, 2009) and late growth of trees (inhibited stem growth (plant height), and promoted lateral growth (crown width and DBH) (Manrique-Alba et al., 2020). The greater the thinning intensity, the more favorable for the growth of trees. Moreover, the increase of forest gap caused by thinning can increase the light and aerodynamic transmission of branches (Pypker et al., 2005). Therefore, it will change the meteorological conditions in the canopy, which control the evaporation of intercepted water (Grunicke et al., 2020). Meanwhile, the asymmetry of tree crown caused by thinning also had a significant impact on tree growth, because it helped trees adapt to light heterogeneity (Niccoli et al., 2020). After thinning, the crown loses symmetry, and the growth ability of the remaining trees reduces due to transverse compression, while in other directions, the width of trees will increase due to larger growth space.

The water used for vegetation mainly comes from the precipitation resources in the area. In some cases, the imbalance between input and output (net consumption) can be so serious that it will permanently destroy the water balance of the ecosystem, leading to soil desiccation (Jia et al., 2017). Soil water is an important factor limiting vegetation growth, controlling vegetation dynamics, and maintaining the ecosystem, especially in arid and semiarid areas (An et al., 2017; Chen et al., 2008). As a common forest management measure, thinning could effectively alleviate this phenomenon. In our study, thinning also effectively increased soil water content. The increasing intensity of thinning was disproportionate to the increase in soil water storage. Specifically, compared with no thinning, the soil water storage of $T_{1/3}$ increased by 41%; $T_{2/3}$ increased by only 57%. Explanations for these results are as follows: high-density forest land consumed a lot of soil water due to high evapotranspiration (Jia et al., 2017; Jian et al., 2015). However, thinning could significantly increase the rainfall reaching the soil surface by changing the above-ground stand structure and reducing rainfall interception (Tague et al., 2019), but also increased the soil evaporation (an increase of exposed area) (del Campo et al., 2014) and individual transpiration (increased availability of light, airflow, and water) (Simonin et al., 2007; Gebhardt et al., 2014), and the competition between roots was reduced, and the water supply was also greatly increased (Sohn et al., 2016). Therefore, the increasing trend of water storage gradually decreases when thinning reaches a certain degree, which is particularly important in the semiarid area (Liu et al., 2018).

With the increase in thinning intensity and the death of residual roots, the remaining trees will absorb more nutrients and water resources by promoting the growth of roots (Krause et al., 2014; Lemay et al., 2018; Olesinski et al., 2012), what is more, improving the photosynthetic capacity of the remaining plants to increase tree growth and water use efficiency (Giuggiola et al., 2013; Hawthorne et al., 2013; Lagergren et al., 2008; Niccoli et al., 2020). Additionally, soil water content is strongly affected by rainfall infiltration and

groundwater recharge (Wu et al., 2021). Recent studies have shown that both root biomass and root characteristics can significantly affect preferential flow and soil water infiltrability (Cui et al., 2022; Huang, Tian, et al., 2017; Jiang et al., 2018; Wu et al., 2021). Root decay increases macropores network density and continuity. Which is very important for the downward movement of soil water. The root channel can promote rainfall infiltration and restore soil water conditions, especially promoting the supplement of deep soil water (Cui et al., 2022; Germann et al., 2012). In our study, we also found that the root decay that follows thinning formed root channels (preferential flow), which enhanced the infiltration rate.

Many researchers have focused on thinning as a strategy to reduce evapotranspiration. Thinning may even lead to a win-win situation, where reducing density can address the impact of water scarcity by: (a) affecting tree death risk, forest health, and productivity and (b) increasing streamflow, usually by reducing forest water demand (Saksa et al., 2017; Tague et al., 2019). The conclusion of this study also just confirmed that moderate thinning ($T_{1/3}$) could effectively adjust the soil water conservation capacity and alleviate soil dryness while maintaining forest production, to achieve a win-win situation. The positive effects of thinning can last for at least 20 years, which can use as a reference to plan further intervention measures (Manrique-Alba et al., 2020; Prescott et al., 2019). Moreover, the thinning method used in this study is a common conventional thinning method (Aboal et al., 2000; Molina & del Campo, 2012), especially in the arid and semiarid regions (Cheng et al., 2020), and the positive effects of this thinning are generally applicable in the context of global climate change and greening Earth (Gebhardt et al., 2014).

5 | CONCLUSIONS

Thinning had a significant impact on the aboveground three-dimensional structure of planted forests, increasing forest gaps ($T_{1/3}$, 17.0%, $T_{2/3}$, 38.0%) and reducing aboveground biomass ($T_{1/3}$, -56.3%, $T_{2/3}$ -75.3%). Moreover, as the above-ground three-dimensional structure changes, the rainfall redistribution pattern had also changed significantly. Thinning increased the penetration of rain ($T_{1/3}$, 43.3%, $T_{2/3}$, 77.5%) and stemflow of forest land ($T_{1/3}$, 44.3%, $T_{2/3}$, 71.0%), and increased the soil water replenishment of forest land. Meanwhile, the withered roots left in the ground rot from root channels, which effectively increased the water infiltration rate, increased groundwater replenishment and effectively relieved soil water deficit. Through trade-off analysis, it was found that non-thinning (NT) yields higher aboveground biomass, and heavy thinning ($T_{2/3}$) yields higher water yield, while moderate thinning ($T_{1/3}$) could maintain certain aboveground biomass while maintaining better soil water content. Therefore, moderate thinning ($T_{1/3}$) could effectively adjust the soil water conservation capacity and alleviate soil dryness without affecting the production of planted forestlands. Under the trend of global climate change and global greening, thinning is a positive effective silvicultural measure.

AUTHOR CONTRIBUTIONS

Ling-Chao Meng: Data curation, formal analysis, investigation, writing original draft. Zeng Cui: data curation, formal analysis, investigation, writing - review and editing. Ze Huang: data curation, writing - review & editing. David Dunkerley: writing - review and editing, supervision. Manuel Esteban Lucas-Borja: writing - review and editing. Gao-Lin Wu: conceptualization, methodology, funding acquisition, investigation, project administration, software, writing - review and editing, supervision.

ACKNOWLEDGMENTS

We thank the editors and anonymous reviewers for their constructive comments and suggestions on this manuscript. This research was funded by the National Natural Science Foundation of China (NSFC41977063, NSFC41930755, NSFC41722107), the Strategic Priority Research Program of the Chinese Academy of Sciences (XDB40000000).

CONFLICT OF INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

DATA AVAILABILITY STATEMENT

All data used for this manuscript are included in the manuscript file.

ORCID

Manuel Esteban Lucas-Borja  <https://orcid.org/0000-0001-6270-8408>

Gao-Lin Wu  <https://orcid.org/0000-0002-5449-7134>

REFERENCES

- Aboal, J. R., Jimenez, M. S., Morales, D., & Gil, P. (2000). Effects of thinning on throughfall in Canary Islands pine forest - the role of fog. *Journal of Hydrology*, 238, 218–230. [https://doi.org/10.1016/S0022-1694\(00\)00329-2](https://doi.org/10.1016/S0022-1694(00)00329-2)
- An, W., Li, Z., Wang, S., Wu, X., Lu, Y., Liu, G., & Fu, B. (2017). Exploring the effects of the "Grain for green" program on the differences in soil water in the semi-arid loess plateau of China. *Ecological Engineering*, 107, 144–151. <https://doi.org/10.1016/j.ecoleng.2017.07.017>
- Bogner, C., Gaul, D., Kolb, A., Schmiedinger, I., & Huwe, B. (2010). Investigating flow mechanisms in forest soil by mixed-effects modeling. *European Journal of Soil Science*, 61, 1079–1090. <https://doi.org/10.1111/j.1365-2389.2010.01300.x>
- Bradford, J. B., & D'Amato, A. W. (2012). Recognizing trade-offs in multi-objective land management. *Frontiers in Ecology and the Environment*, 10, 210–216. <https://doi.org/10.1890/110031>
- Cao, S., Chen, L., Shankman, D., Wang, C., Wang, X., & Zhang, H. (2011). Excessive reliance on afforestation in China's arid and semi-arid regions: Lessons in ecological restoration. *Earth-Science Reviews*, 104, 240–245. <https://doi.org/10.1016/j.earscirev.2010.11.002>
- Chen, H.S., Shao, M.A., & Li, Y.Y. (2008). Soil desiccation in the Loess Plateau of China. *Geoderma*, 143, 91–100. <https://doi.org/10.1016/j.geoderma.2007.10.013>
- Cheng, X., Bai, Y., Zhu, J., & Han, H. (2020). Effects of forest thinning on interception and surface runoff in *Larix principis-rupprechtii* plantation during the growing season. *Journal of Arid Environments*, 181, 104222. <https://doi.org/10.1016/j.jaridenv.2020.104222>
- Chirino, E., Bonet, A., Bellot, J., & Sanchez, J. R. (2006). Effects of 30-year-old Aleppo pine plantations on runoff, soil erosion, and plant diversity in a semi-arid landscape in south eastern Spain. *Catena*, 65, 19–29. <https://doi.org/10.1016/j.catena.2005.09.003>
- Cui, Z., Huang, Z., Liu, Y., Lopez-Vicente, M., & Wu, G. L. (2022). Natural compensation mechanism of soil water infiltration through decayed roots in semi-arid vegetation species. *Science of the Total Environment*, 819, 151985. <https://doi.org/10.1016/j.scitotenv.2021.151985>
- del Campo, A. D., Fernandes, T. J. G., & Molina, A. J. (2014). Hydrology-oriented (adaptive) silviculture in a semiarid pine plantation: How much can be modified the water cycle through forest management? *European Journal of Forest Research*, 133, 879–894. <https://doi.org/10.1007/s10342-014-0805-7>
- del Campo, A. D., Gonzalez-Sanchis, M., Molina, A. J., Garcia-Prats, A., Ceacero, C. J., & Bautista, I. (2019). Effectiveness of water-oriented thinning in two semiarid forests: The redistribution of increased net rainfall into soil water, drainage and runoff. *Forest Ecology and Management*, 438, 163–175. <https://doi.org/10.1016/j.foreco.2019.02.020>
- Deng, L., Yan, W., Zhang, Y., & Shangguan, Z. P. (2016). Severe depletion of soil moisture following land-use changes for ecological restoration: Evidence from northern China. *Forest Ecology and Management*, 366, 1–10. <https://doi.org/10.1016/j.foreco.2016.01.026>
- Dung, B. X., Gomi, T., Miyata, S., Sidle, R. C., Kosugi, K., & Onda, Y. (2012). Runoff responses to forest thinning at plot and catchment scales in a headwater catchment draining Japanese cypress forest. *Journal of Hydrology*, 444, 51–62. <https://doi.org/10.1016/j.jhydrol.2012.03.040>
- Elbakidze, M., Angelstam, P., Andersson, K., Nordberg, M., & Pautov, Y. (2011). How does forest certification contribute to boreal biodiversity conservation? Standards and outcomes in Sweden and NW Russia. *Forest Ecology and Management*, 262, 1983–1995. <https://doi.org/10.1016/j.foreco.2011.08.040>
- FAO. (2015). *Global Forest Resources Assessment 2015* (FAO Forestry Paper No. 1). Rome, Italy: UN Food and Agriculture Organization. [http://refhub.elsevier.com/S0378-1127\(18\)31510-X/h0085](http://refhub.elsevier.com/S0378-1127(18)31510-X/h0085)
- Feng, X., Fu, B., Piao, S., Wang, S., Ciais, P., Zeng, Z., Lu, Y., Zeng, Y., Li, Y., Jiang, X., & Wu, B. (2016). Revegetation in China's Loess Plateau is approaching sustainable water resource limits. *Nature Climate Change*, 6, 1019–1022. <https://doi.org/10.1038/nclimate3092>
- Feng, X., Fu, B., Zhang, Y., Pan, N., Zeng, Z., Tian, H., Lyu, Y., Chen, Y., Ciais, P., Wang, Y., Zhang, L., Cheng, L., Maestre, F. T., Fernández-Martínez, M., Sardans, J., & Peuelas, J. (2021). Recent leveling off of vegetation greenness and primary production reveals the increasing soil water limitations on the greening Earth. *Science Bulletin*, 66, 1462–1471. <https://doi.org/10.1016/j.scib.2021.02.023>
- Gauthier, M. M., & Jacobs, D. F. (2009). Short-term physiological responses of black walnut (*Juglans nigra* L.) to plantation thinning. *Forest Science*, 55, 221–229. <https://doi.org/10.1093/forestscience/55.3.221>
- Ge, J., Pitman, A. J., Guo, W., Zan, B., & Fu, C. (2020). Impact of revegetation of the loess plateau of China on the regional growing season water balance. *Hydrology and Earth System Sciences*, 24, 515–533. <https://doi.org/10.5194/hess-24-515-2020>
- Gebhardt, T., Haerberle, K. H., Matyssek, R., Schulz, C., & Ammer, C. (2014). The more, the better? Water relations of Norway spruce stands after progressive thinning. *Agricultural and Forest Meteorology*, 197, 235–243. <https://doi.org/10.1016/j.agrformet.2014.05.013>
- Germann, P.F., Lange, B., & Lüscher, P. (2012). Preferential flow dynamics and plant root systems. *Hydrogeology*, 121–141. <https://doi.org/10.1016/B978-0-12-386941-8.00004-6>
- Gerwing, J.J., & Farias, D.L. (2000). Integrating liana abundance and forest stature into an estimate of total aboveground biomass for an eastern Amazonian forest. *Journal of Tropical Ecology*, 16, 327–335. <https://doi.org/10.1017/S0266467400001437>
- Grunicke, S., Queck, R., & Bernhofer, C. (2020). Long-term investigation of forest canopy rainfall interception for a spruce stand. *Agricultural and*

- Forest Meteorology, 292, 108125. <https://doi.org/10.1016/j.agrformet.2020.108125>
- Giuggiola, A., Bugmann, H., Zingg, A., Dobbertin, M., & Rigling, A. (2013). Reduction of stand density increases drought resistance in xeric scots pine forests. *Forest Ecology and Management*, 310, 827–835. <https://doi.org/10.1016/j.foreco.2013.09.030>
- Guillemot, J., Klein, E. K., Davi, H., & Courbet, F. (2015). The effects of thinning intensity and tree size on the growth response to annual climate in *Cedrus atlantica*: A linear mixed modeling approach. *Annals of Forest Science*, 72, 651–663. <https://doi.org/10.1007/s13595-015-0464-y>
- Hawthorne, S. N. D., Lane, P. N. J., Bren, L. J., & Sims, N. C. (2013). The long-term effects of thinning treatments on vegetation structure and water yield. *Forest Ecology and Management*, 310, 983–993. <https://doi.org/10.1016/j.foreco.2013.09.046>
- Huang, J., Yu, H., Dai, A., Wei, Y., & Kang, L. (2017). Drylands face potential threat under 2 °C global warming target. *Nature Climate Change*, 7, 417–422. <https://doi.org/10.1038/nclimate3275>
- Huang, Z., Tian, F. P., Wu, G. L., Liu, Y., & Dang, Z. Q. (2017). Legume grasslands promote precipitation infiltration better than Gramineous grasslands in arid regions. *Land Degradation & Development*, 28, 309–316. <https://doi.org/10.1002/ldr.2635>
- Huang, Z., Yang, W. J., Liu, Y., Shen, W. B., López-Vicente, M., & Wu, G. L. (2020). Belowground soil water response in the afforestation-cropland interface under semi-arid conditions. *Catena*, 193, 104660. <https://doi.org/10.1016/j.catena.2020.104660>
- IPCC. (2014). *Fifth assessment report, climate change 2014: Synthesis report*. Cambridge, UK: Cambridge University Press. [http://refhub.elsevier.com/S0378-1127\(18\)30150-6/h0095](http://refhub.elsevier.com/S0378-1127(18)30150-6/h0095)
- Jia, X. X., Shao, M. A., Zhu, Y., & Luo, Y. (2017). Soil moisture decline due to afforestation across the Loess Plateau, China. *Journal of Hydrology*, 546, 113–122. <https://doi.org/10.1016/j.jhydrol.2017.01.011>
- Jian, S., Zhao, C., Fang, S., & Yu, K. (2015). Effects of different vegetation restoration on soil water storage and water balance in the Chinese-Loess Plateau. *Agricultural and Forest Meteorology*, 206, 85–96. <https://doi.org/10.1016/j.agrformet.2015.03.009>
- Jiang, X. J., Liu, W., Chen, C., Liu, J., Yuan, Z. Q., Jin, B., & Yu, X. (2018). Effects of three morphometric features of roots on soil water flow behavior in three sites in China. *Geoderma*, 320, 161–171. <https://doi.org/10.1016/j.geoderma.2018.01.035>
- Jimenez, E., Vega, J. A., Fernandez, C., & Fonturbel, T. (2011). Is pre-commercial thinning compatible with carbon sequestration? A case study in a maritime pine stand in northwestern Spain. *Forestry*, 84, 149–157. <https://doi.org/10.1093/forestry/cpr002>
- Krause, C., Lemay, A., Tremblay, S., Ruel, J. C., & Plourde, P. Y. (2014). How does the root system inhibit windthrow in thinned black spruce sites in the boreal forest? *Trees-Structure and Function*, 28, 1723–1735. <https://doi.org/10.1007/s00468-014-1080-4>
- Lagergren, F., Lankreijer, H., Kucera, J., Cienciala, E., Moelder, M., & Lindroth, A. (2008). Thinning effects on pine-spruce forest transpiration in Central Sweden. *Forest Ecology and Management*, 255, 2312–2323. <https://doi.org/10.1016/j.foreco.2007.12.047>
- Legendre, P., Oksanen, J., & ter Braak, C. J. F. (2011). Testing the significance of canonical axes in redundancy analysis. *Methods in Ecology and Evolution*, 2, 269–277. <https://doi.org/10.1111/j.2041-210X.2010.00078.x>
- Lemay, A., Krause, C., & Achim, A. (2018). Comparison of wood density in roots and stems of black spruce before and after commercial thinning. *Forest Ecology and Management*, 408, 94–102. <https://doi.org/10.1016/j.foreco.2017.10.042>
- Levia, D. F., & Frost, E. E. (2003). A review and evaluation of stemflow literature in the hydrologic and biogeochemical cycles of forested and agricultural ecosystems. *Journal of Hydrology*, 274, 1–29. [https://doi.org/10.1016/S0022-1694\(02\)00399-2](https://doi.org/10.1016/S0022-1694(02)00399-2)
- Limousin, J. M., Rambal, S., Ourcival, J. M., & Joffre, R. (2008). Modelling rainfall interception in a Mediterranean *Quercus ilex* ecosystem: Lesson from a throughfall exclusion experiment. *Journal of Hydrology*, 357, 57–66. <https://doi.org/10.1016/j.jhydrol.2008.05.001>
- Liu, Y., Miao, H. T., Huang, Z., Cui, Z., He, H., Zheng, J., Han, F., Chang, X., & Wu, G. L. (2018). Soil water depletion patterns of artificial forest species and ages on the Loess Plateau (China). *Forest Ecology and Management*, 417, 137–143. <https://doi.org/10.1016/j.foreco.2018.03.005>
- Lu, N., Fu, B., Jin, T., & Chang, R. (2014). Trade-off analyses of multiple ecosystem services by plantations along a precipitation gradient across Loess Plateau landscapes. *Landscape Ecology*, 29, 1697–1708. <https://doi.org/10.1007/s10980-014-0101-4>
- Manrique-Alba, A., Begueria, S., Molina, A. J., Gonzalez-Sanchis, M., Tomas-Burguera, M., del Campo, A. D., Colangelo, M., & Camarero, J. J. (2020). Long-term thinning effects on tree growth, drought response and water use efficiency at two Aleppo pine plantations in Spain. *Science of the Total Environment*, 728, 138536. <https://doi.org/10.1016/j.scitotenv.2020.138536>
- Molina, A. J., & del Campo, A. D. (2012). The effects of experimental thinning on throughfall and stemflow: A contribution towards hydrology-oriented silviculture in Aleppo pine plantations. *Forest Ecology and Management*, 269, 206–213. <https://doi.org/10.1016/j.foreco.2011.12.037>
- Niccoli, F., Pelleri, F., Manetti, M. C., Sansone, D., & Battipaglia, G. (2020). Effects of thinning intensity on productivity and water use efficiency of *Quercus robur* L. *Forest Ecology and Management*, 473, 118282. <https://doi.org/10.1016/j.foreco.2020.118282>
- Olesinski, J., Lavigne, M. B., Kershaw, J. A., Jr., & Krasowski, M. J. (2012). Fine-root dynamics change during stand development and in response to thinning in balsam fir (*Abies balsamea* L. mill.) forests. *Forest Ecology and Management*, 286, 48–58. <https://doi.org/10.1016/j.foreco.2012.08.047>
- Olivar, J., Bogino, S., Rathgeber, C., Bonnesoeur, V., & Bravo, F. (2014). Thinning has a positive effect on growth dynamics and growth-climate relationships in Aleppo pine (*Pinus halepensis*) trees of different crown classes. *Annals of Forest Science*, 71, 395–404. <https://doi.org/10.1007/s13595-013-0348-y>
- Park, J., Kim, T., Moon, M., Cho, S., Ryu, D., & Kim, H. S. (2018). Effects of thinning intensities on tree water use, growth, and resultant water use efficiency of 50-year-old *Pinus koraiensis* forest over four years. *Forest Ecology and Management*, 408, 121–128. <https://doi.org/10.1016/j.foreco.2017.09.031>
- Prescott, C., deMontigny, L., & Harper, G. (2019). Eighteen-year growth responses to thinning and fertilization of a height-repressed lodgepole pine stand in interior British Columbia. *Forestry Chronicle*, 95, 207–221. <https://doi.org/10.5558/tfc2019-029>
- Pypker, T. G., Bond, B. J., Link, T. E., Marks, D., & Unsworth, M. H. (2005). The importance of canopy structure in controlling the interception loss of rainfall: Examples from a young and an old-growth Douglas-fir forest. *Agricultural and Forest Meteorology*, 130, 113–129. <https://doi.org/10.1016/j.agrformet.2005.03.003>
- Saksa, P. C., Conklin, M. H., Battles, J. J., Tague, C. L., & Bales, R. C. (2017). Forest thinning impacts on the water balance of Sierra Nevada mixed-conifer headwater basins. *Water Resources Research*, 53, 5364–5381. <https://doi.org/10.1002/2016wr019240>
- Simonin, K., Kolb, T. E., Montes-Helu, M., & Koch, G. W. (2007). The influence of thinning on components of stand water balance in a ponderosa pine forest stand during and after extreme drought. *Agricultural and Forest Meteorology*, 143, 266–276. <https://doi.org/10.1016/j.agrformet.2007.01.003>
- Sohn, J. A., Saha, S., & Bauhus, J. (2016). Potential of forest thinning to mitigate drought stress: A meta-analysis. *Forest Ecology and Management*, 380, 261–273. <https://doi.org/10.1016/j.foreco.2016.07.046>

- Stanturf, J. A., & Mansourian, S. (2020). Forest landscape restoration: State of play. *Royal Society Open Science*, 7, 201218. <https://doi.org/10.1098/rsos.201218>
- Sun, X. C., Onda, Y. C., Kato, H., Gomi, T., & Komatsu, H. (2015). Effect of strip thinning on rainfall interception in a Japanese cypress plantation. *Journal of Hydrology*, 525, 607–618. <https://doi.org/10.1016/j.jhydrol.2015.04.023>
- Sun, X. C., Onda, Y. C., & Kato, H. (2014). Incident rainfall partitioning and canopy interception modeling for an abandoned Japanese cypress stand. *Journal of Forest Research*, 19, 317–328. <https://doi.org/10.1007/s10310-013-0421-2>
- Swaffer, B. A., Holland, K. L., Doody, T. M., & Hutson, J. (2014). Rainfall partitioning, tree form and measurement scale: A comparison of two co-occurring, morphologically distinct tree species in a semi-arid environment. *Ecohydrology*, 7, 1331–1344. <https://doi.org/10.1002/eco.1461>
- Tague, C. L., Moritz, M., & Hanan, E. (2019). The changing water cycle: The eco-hydrologic impacts of forest density reduction in mediterranean (seasonally dry) regions. *Wiley Interdisciplinary Reviews-Water*, 6(4), e1350. <https://doi.org/10.1002/wat2.1350>
- Wang, X. M., Zhang, C. X., Hasi, E., & Dong, Z. B. (2010). Has the three norths Forest shelterbelt program solved the desertification and dust storm problems in arid and semiarid China? *Journal of Arid Environments*, 74, 13–22. <https://doi.org/10.1016/j.jaridenv.2009.08.001>
- Wu, G. L., Huang, Z., Liu, Y. F., Cui, Z., Liu, Y., Chang, X. F., Tian, F. P., Lopez-Vicente, M., & Shi, Z. H. (2019). Soil water response of plant functional groups along an artificial legume grassland succession under semi-arid conditions. *Agricultural and Forest Meteorology*, 278, 107670. <https://doi.org/10.1016/j.agrformet.2019.107670>
- Wu, G. L., Cui, Z., & Huang, Z. (2021). Contribution of root decay process on soil infiltration capacity and soil water replenishment of planted forestland in semi-arid regions. *Geoderma*, 404, 115289. <https://doi.org/10.1016/j.geoderma.2021.115289>
- Wu, G. L., Yang, Z., Cui, Z., Liu, Y., Fang, N. F., & Shi, Z. H. (2016). Mixed artificial grasslands with more roots improved mine soil infiltration capacity. *Journal of Hydrology*, 535, 54–60. <https://doi.org/10.1016/j.jhydrol.2016.01.059>
- Zastrow, M. (2019). China's tree-planting could falter in a warming world. *Nature*, 573, 474–475. <https://doi.org/10.1038/d41586-019-02789-w>

How to cite this article: Meng, L.-C., Cui, Z., Huang, Z., Lucas-Borja, M. E., Dunkerley, D., & Wu, G.-L. (2022). Thinning effects on forest production and rainfall redistribution: Reduced soil water deficit and improved sustainability of semiarid plantation forestlands. *Land Degradation & Development*, 33(16), 3163–3173. <https://doi.org/10.1002/ldr.4379>